

Human-AI Co-Creation for Workplan Generation in Manufacturing: Optimizing Tool selection and Clamping strategies through Human-AI Dialogue

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Thesis submitted for the degree of
Master of Science in Mechanical
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Preface

This is my acknowledgment to thank everyone who has kept me busy. I would like to thank my supervisor, my advisor, and the entire jury. My family has also been very supportive, of course.

Jordaan Dimitrov

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Abstract

This **abstract** environment provides a summary of the work, which may or may not be extensive. The intention is that this should be limited to 1 page.

- Autocam context, fixture planning
- human expertise dependence, brittle rule based systems
- aim is AI assisted clamping region detection
- method used is supervised learning, feedforward NN, face/mesh level processing
- training data obtained via manual labeling and parametric part generation
- processing done in multiple ways, cad face extraction, discretized mesh, adaptive mesh refinement
- surface/mesh information fed into NN: area; normals; outerness; opposing-surface validation; wall thickness
- constraints are physical feasibility, reachability, total clamping force (thin walls)
- heavy computation via HPC cluster, training on gpu nodes, monitoring via 3 data sets (training, validation, xyz), one for tuning hyper parameters
- Metrics: precision; recall; F1-like combined score; false-positive suppression since that is more dangerous than missing a solution
- output gives user multiple solutions, can be ordered based on reachability, stability, etc
- challenges
- future work

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List of abbreviations and symbols

abbreviations

NN	Neural Network
LoG	Laplacian-of-Gaussian
MSE	Mean Square error
PSNR	Peak Signal-to-Noise ratio

Symbols

42	“The Answer to the Ultimate Question of Life, the Universe, and Everything” volgens de [1]
c	Speed of light
E	Energy
m	Mass
π	The number pi

Chapter 1

Introduction

The first contains a general introduction to the work. The goals are defined and the modus operandi is explained.

1.1 Overview AutoCAM project and thesis context

1.2 Motivation

1.3 Problem statement

1.4 Objectives

Chapter 2

Background and related work

A chapter is a logical unit. It normally starts with an introduction, which you are reading now. The last topic of the chapter holds the conclusion.

2.1 Current Fixture Planning Approaches

First comes the introduction to this topic.

2.1.1 Rule Based Generative fixture design

- yes no decision trees
- hardcoded geometric constraints
- deterministic logic means low flexibility
- brittle with more difficult geometry
- low adaptability

2.1.2 Case based variant fixture design

- Group technology classification (eg opitz)
- case library of prior part solutions
- similarity search
- adaptation step
- dataset quality determines performance
- limited innovation because biased toward historical design, unless adaptation step is very sophisticated
- adaptation hard for new features
- schematic [2.2](#)

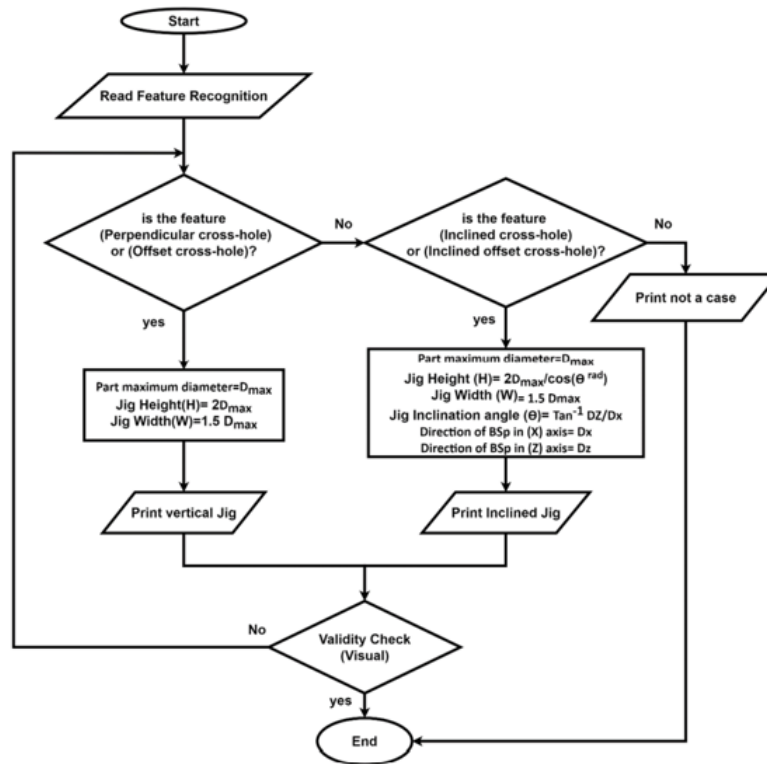


Figure 5. Flowchart of computer-aided traditional jigs and fixtures design.

FIGURE 2.1: Generative fixture design decision tree (source)

2.2 AI in manufacturing

2.2.1 General role of AI in manufacturing

- data driven optimization
- reducing human dependency

2.2.2 AI for CAD/CAM applications

- feature/pattern recognition
- automatic toolpath generation
- geometric classification (paper from presentation) [2.3](#)

2.2.3 Motivation for AI based methods

- abstraction beyond hard coded rules

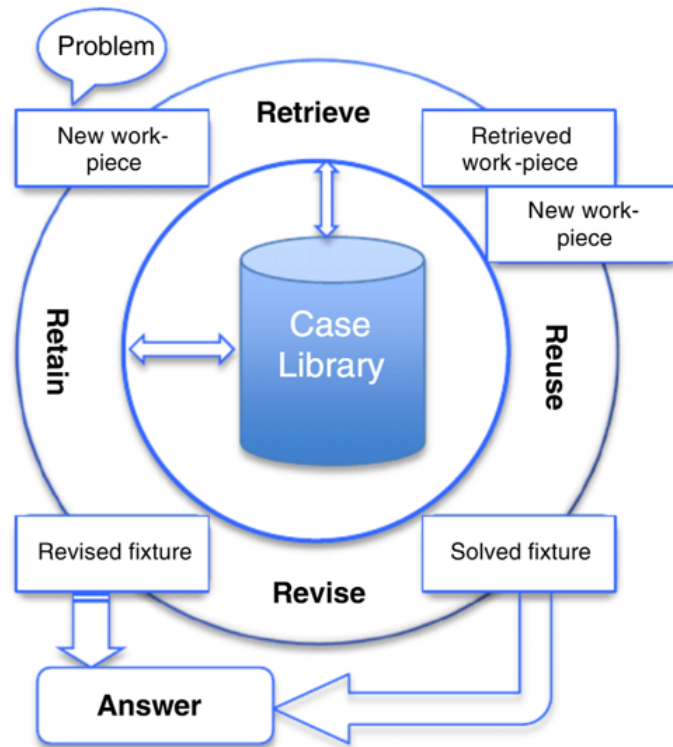


FIGURE 2.2: Variant fixture design scheme (source)

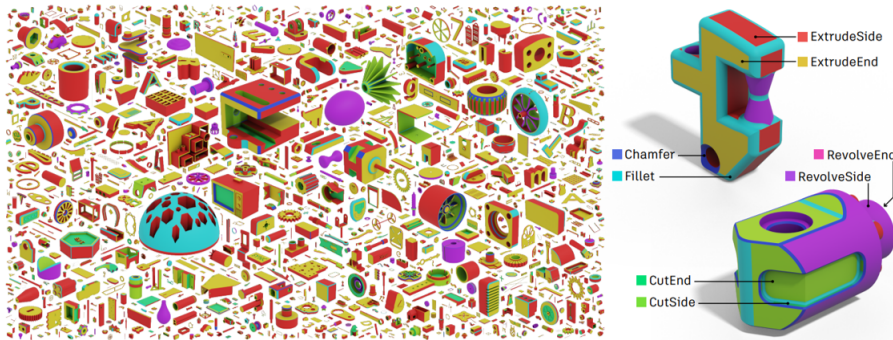


Figure 4: An overview of 3D models from the Fusion 360 Gallery segmentation dataset (left). Each 3D model is labeled according to the CAD modeling operations used to create it (right).

FIGURE 2.3: Deep learning classification

- ability to handle many geometric inputs without breaking (robustness)
- probabilistic prediction

2.2.4 Challenges & Limitations

- large labeled dataset needed
- computational cost
- explainability issues (black box)
- safety concerns with decision making (liability?)

2.2.5 Trends in Industry

- Move toward hybrid human-AI system, rather than fully autonomous decision making
- digital twins and AI monitoring
- predictive maintenance

2.3 Different CAD representation models

- BREP
- voxel
- mesh (stl)
- point cloud
- back this up with literature and give pros and cons of each, i can even reference the virtual product development course

2.4 Clamping methods

2.4.1 Straight Vise

2.4.2 Double V block

2.4.3 V block + flat

2.5 Conclusion

- classical methods rigid, brittle and have limited adaptability
- ai enables datadriven reasoning, robustness, abstraction, probabilistic outputs

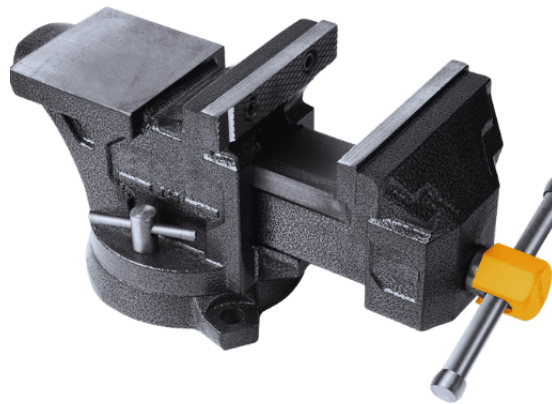


FIGURE 2.4: Vise



FIGURE 2.5: V block double

- but has its challenges such as data requirements, computational cost, explainability, safety
- industry trend: hybrid workflow



FIGURE 2.6: V block + Flat

Chapter 3

Methodology

3.1 Surface processing pipeline

- Cad model is input
- extraction surfaces
- filtering out irrelevant geometry?
- computation of geometric descriptors (normals, curvature, area etc)

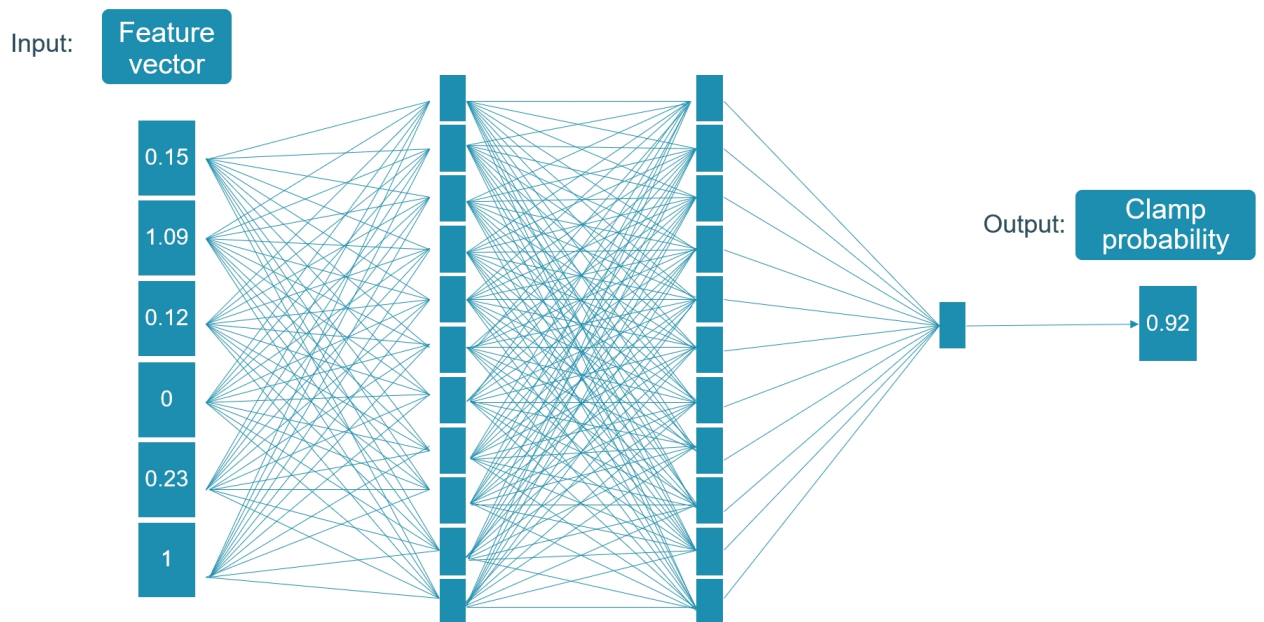


FIGURE 3.1: Feedforward network visualization with two hidden layers

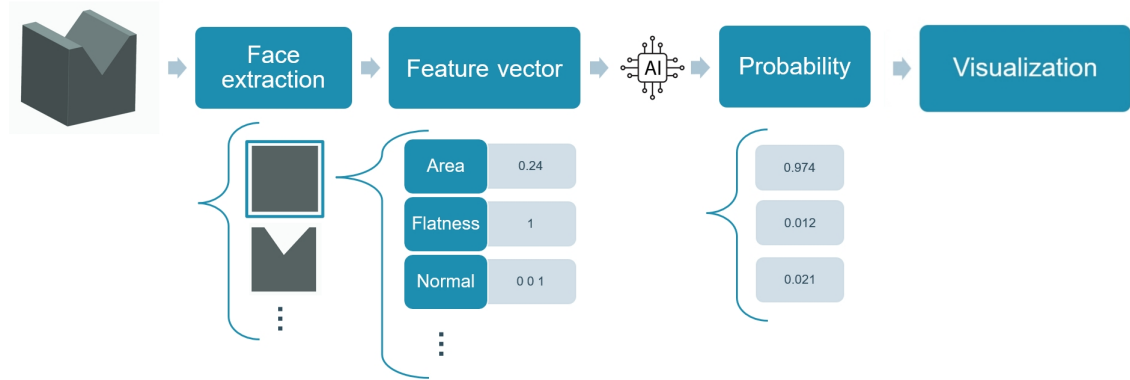


FIGURE 3.2: Pipeline for surface processing

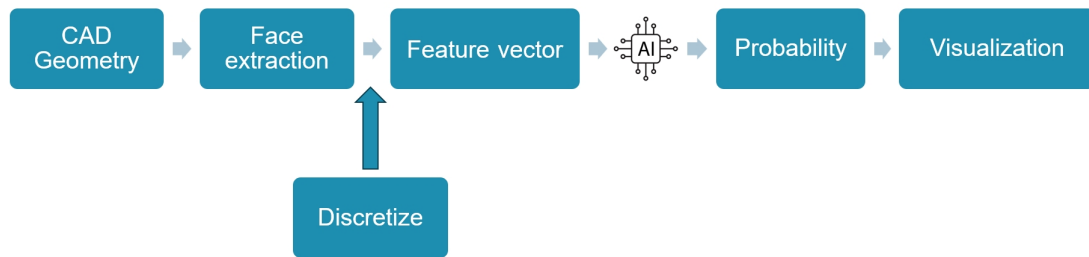


FIGURE 3.3: Discretized Pipeline

3.2 Discretized Geometry Processing

- conversion of brep surfaces into discretized mesh
- adaptive mesh refinement
- computationally much heavier
- physical constraints into loss function: net clamping force zero, patch size constraint to prevent isolated micro triangles
- more uncertainty

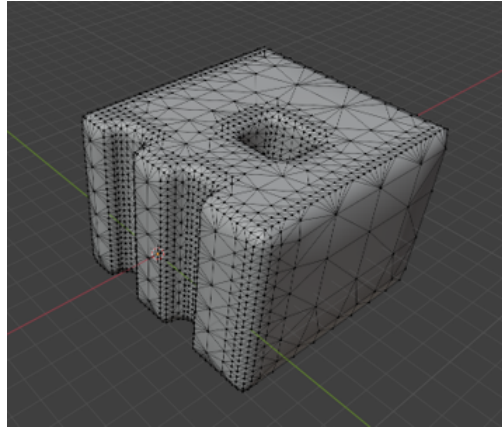


FIGURE 3.4: Adaptive mesh refinement

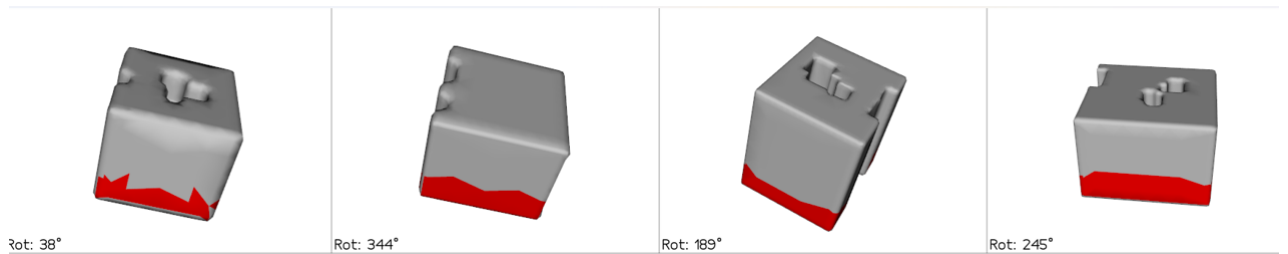


FIGURE 3.5: Parametrically generating parts

3.3 Scaling model

- HPC cluster (reference them as they ask on their website)
- optimize model to run on GPU node (parallelization)
- parametric modeling to create a large number of parts
- used Blender with base part and modifications such as rotation scaling holes chamfers etc

3.4 Feature enrichment

- goal is to capture more geometric information to feed into model, therefore the following features were added

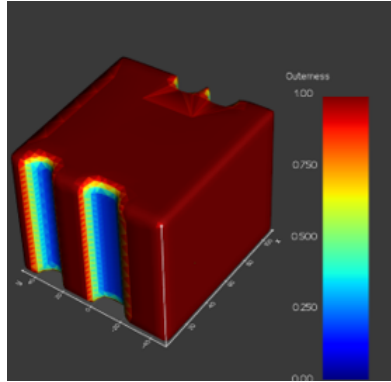


FIGURE 3.6: Outerness

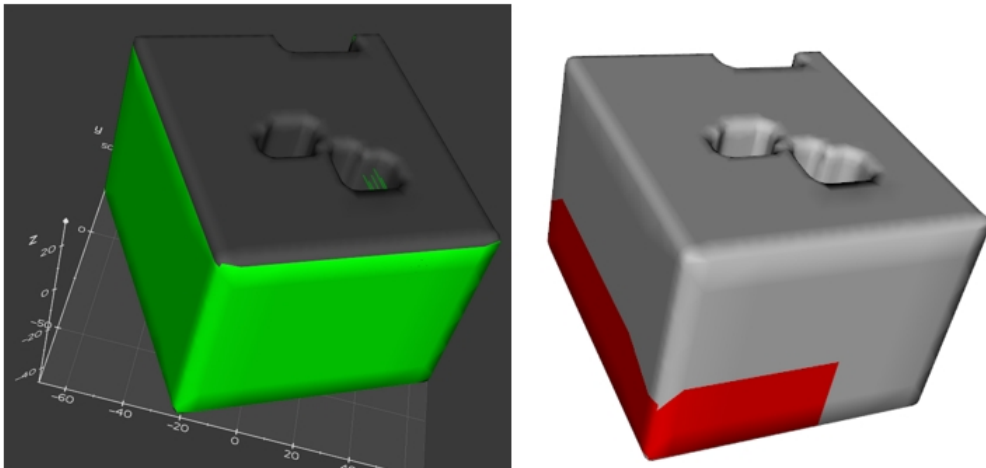


FIGURE 3.7: Opposing surface validation

3.4.1 Outerness

3.4.2 Opposing surface validation

3.4.3 Wall thickness

3.5 Post processing

- AI model outputs a single colored region containing multiple solutions
- goal is to extract the different solutions and rank them based on feasibility, reachability, stability etcand provide a visualization
- The goal is not to automatically choose the best solution, it is a tool to help the process planner or operator by providing a suggestion which he/she should approve

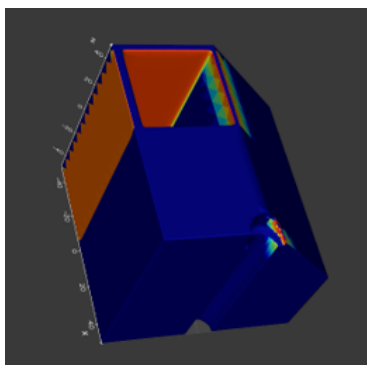


FIGURE 3.8: Wall Thickness

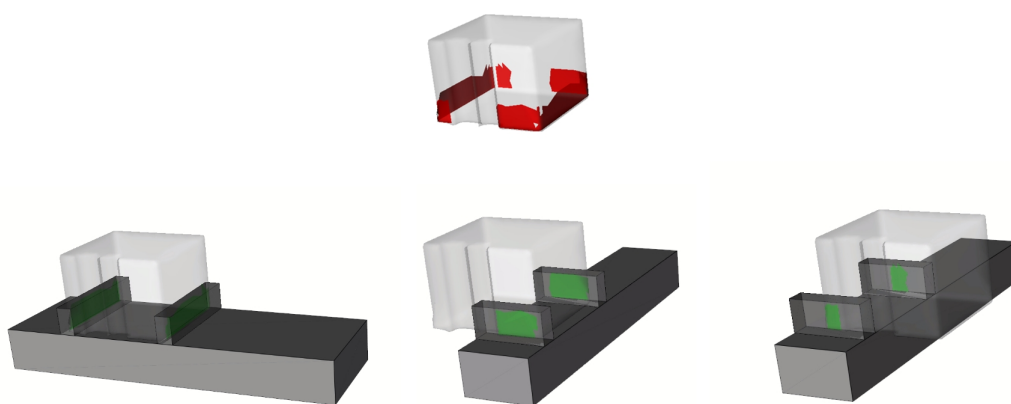


FIGURE 3.9: Visualization of different solutions on the same part in post processing

3.6 Conclusion

Chapter 4

Results

4.1 Surface processing

- Probability heat map
- Overfitting
- Labeling accuracy

4.2 Discretized geometry processing

4.3 Conclusion

4. RESULTS

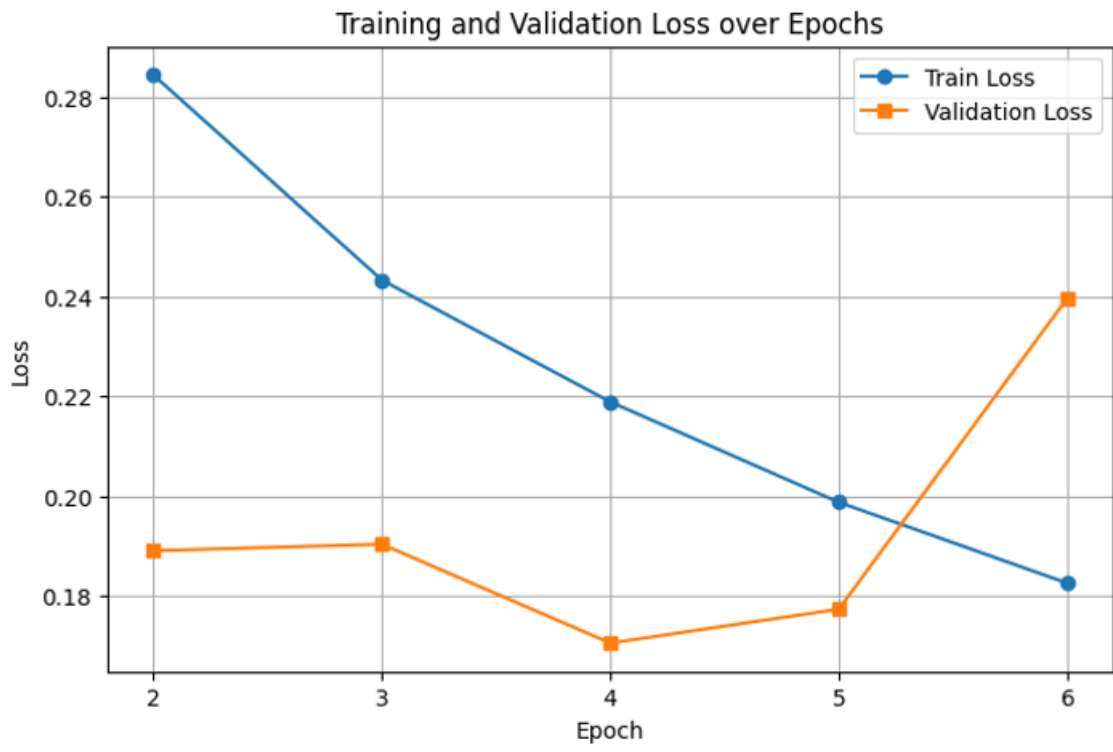


FIGURE 4.1: Overfitting example

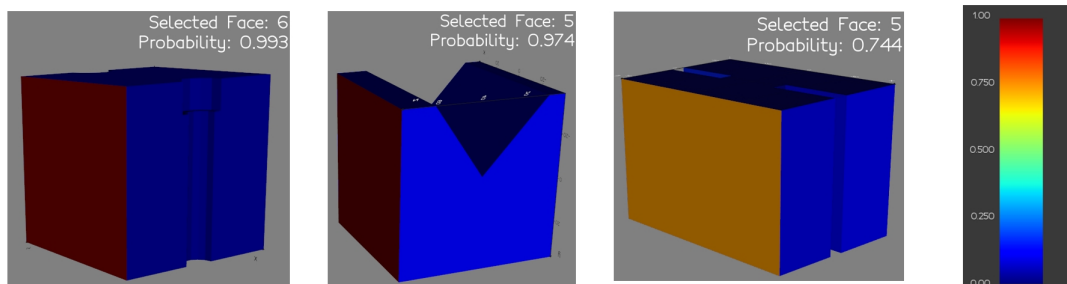


FIGURE 4.2: Visualization examples surface processing

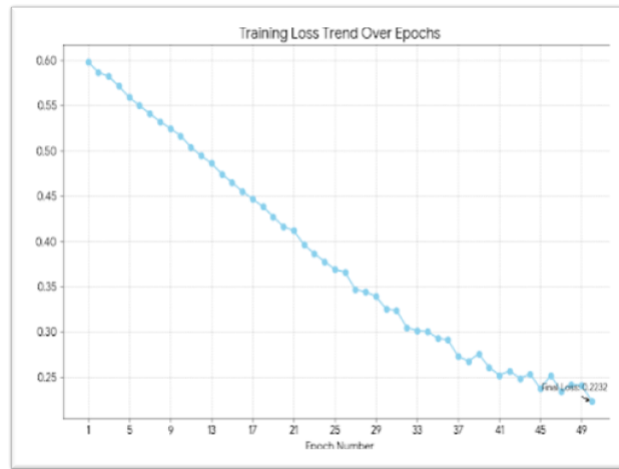


FIGURE 4.3: Visualization examples surface processing

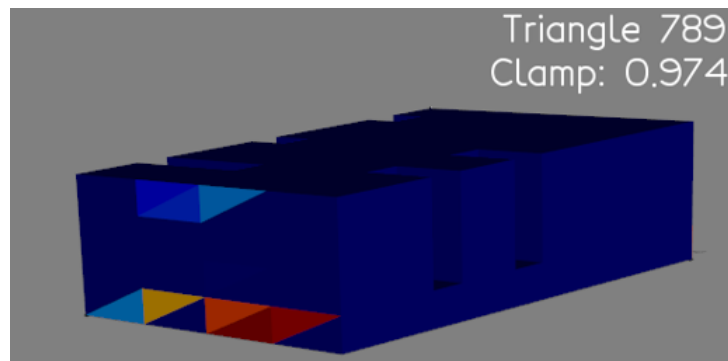


FIGURE 4.4: First result discretized

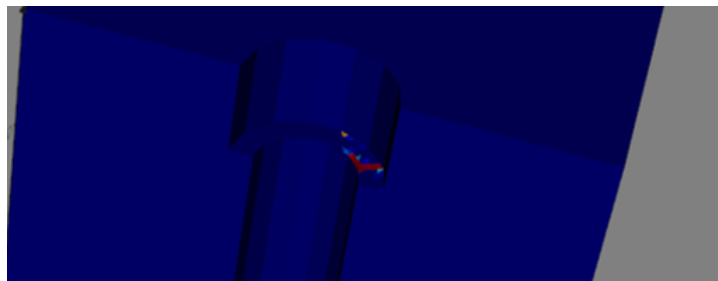


FIGURE 4.5: Illustration of micropatch solution

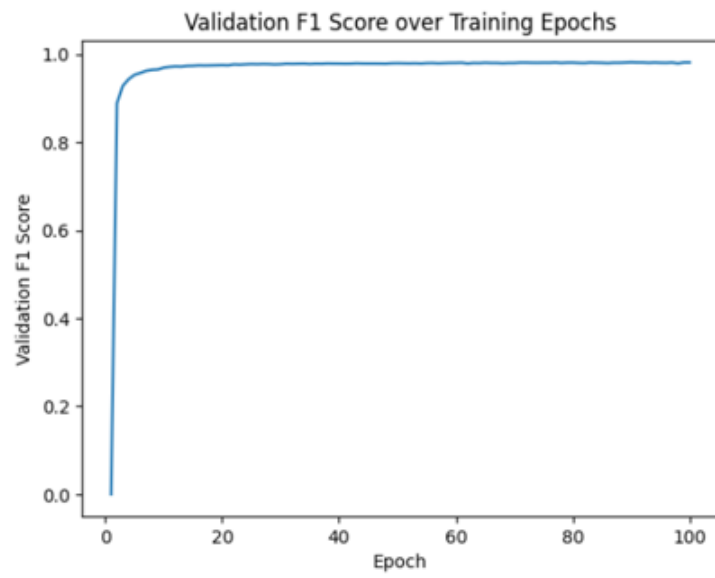


FIGURE 4.6: F1 score result

Chapter 5

Discussion and Future Research Directions

5.1 The First Topic of this Chapter

5.1.1 Item 1

Sub-item 1

Sub-item 2

5.1.2 Item 2

5.2 The Second Topic

5.3 Conclusion

Chapter 6

Conclusion

The final chapter contains the overall conclusion. It also contains suggestions for future work and industrial applications.

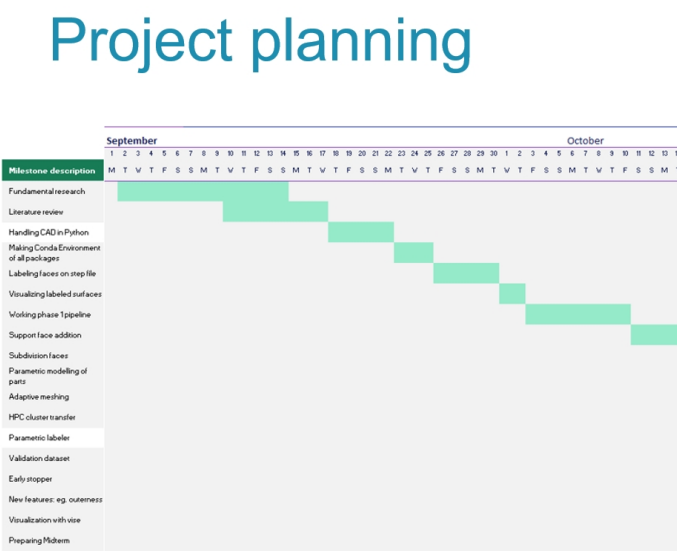
Appendices

Appendix A

The First Appendix

Appendices hold useful data which is not essential to understand the work done in the master’s thesis. An example is a (program) source. An appendix can also have sections as well as figures and references[1].

A.1 Gant chart



70.3770.377

FIGURE A.1: Gant chart

Appendix B

The Last Appendix

Appendices are numbered with letters, but the sections and subsections use arabic numerals, as can be seen below.

B.1 Lorem 20-24

B.2 Lorem 25-27

Bibliography

- [1] D. Adams. *The Hitchhiker's Guide to the Galaxy*. Del Rey (reprint), 1995. ISBN-13: 978-0345391803.
- [2] T. Pratchett and N. Gaiman. *Good Omens: The Nice and Accurate Prophecies of Agnes Nutter, Witch*. HarperTorch (reprint), 2006. ISBN-13: 978-0060853983.
- [3] Wikipedia. Thesis or dissertation. URL: http://en.wikipedia.org/wiki/Thesis_or_dissertation, last checked on 2010-01-07.

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